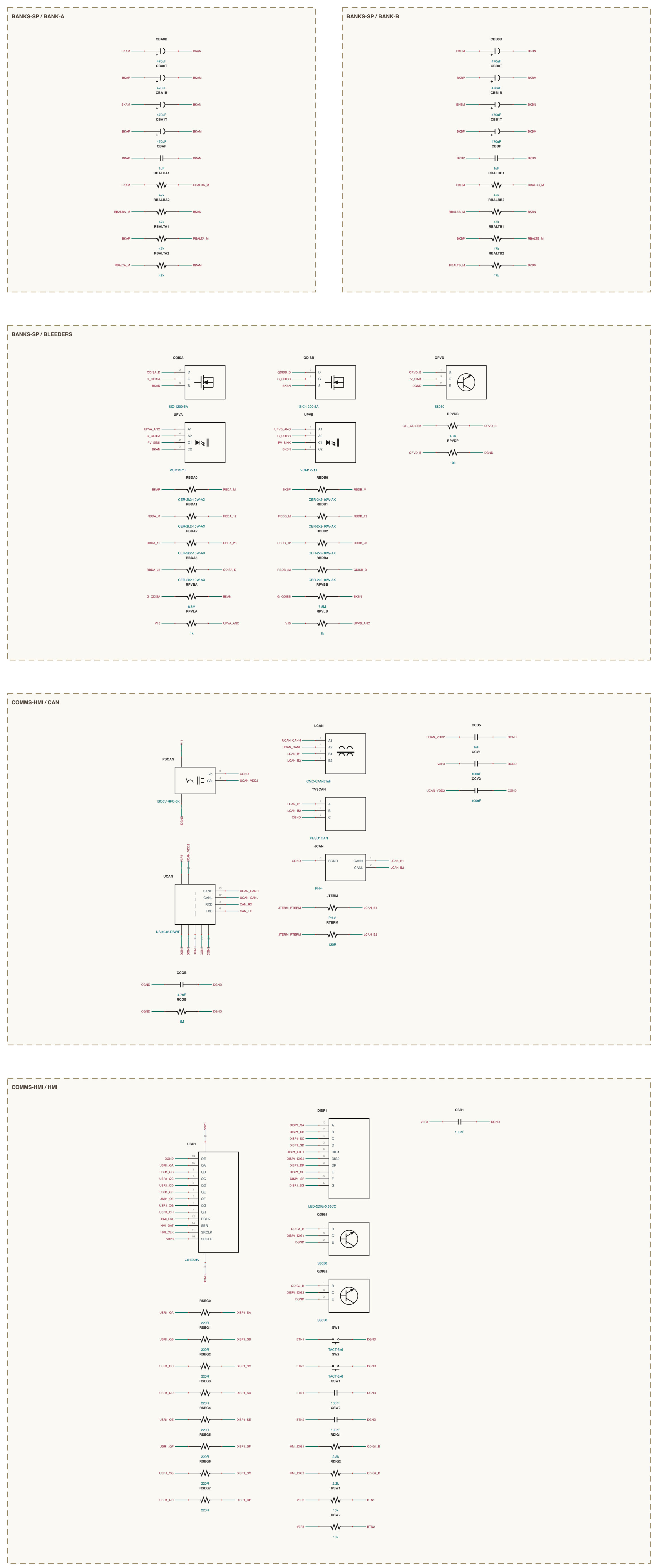
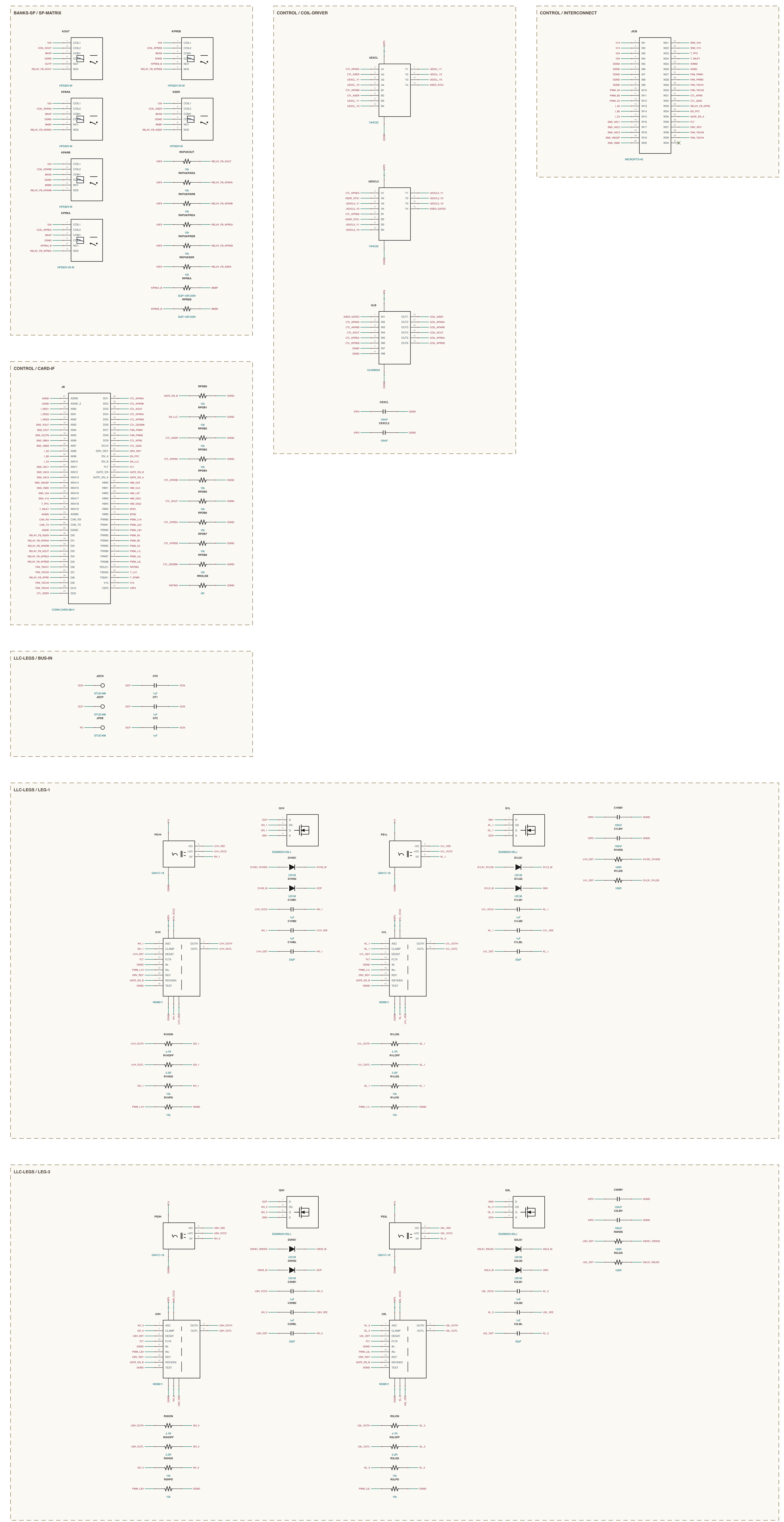
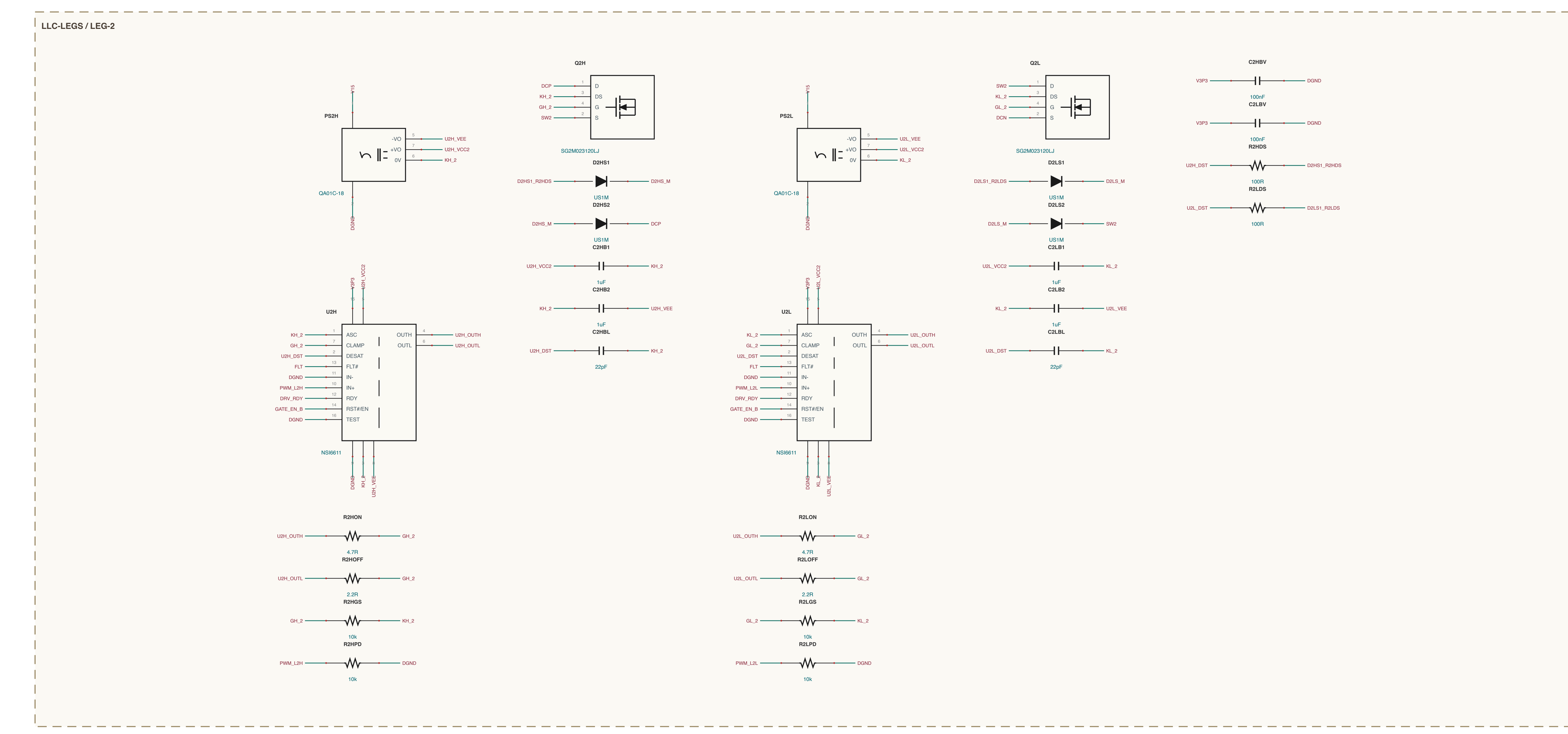
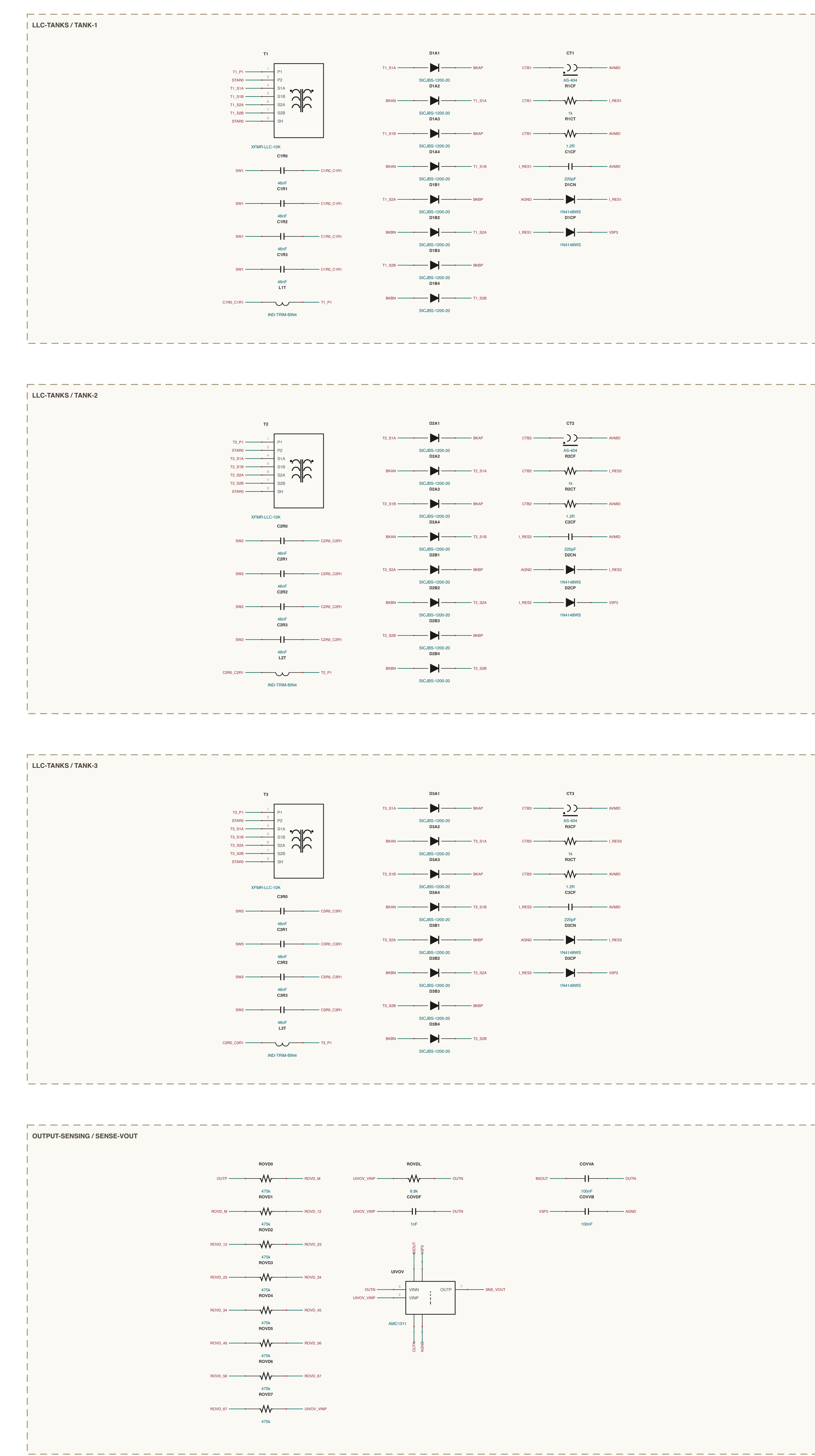
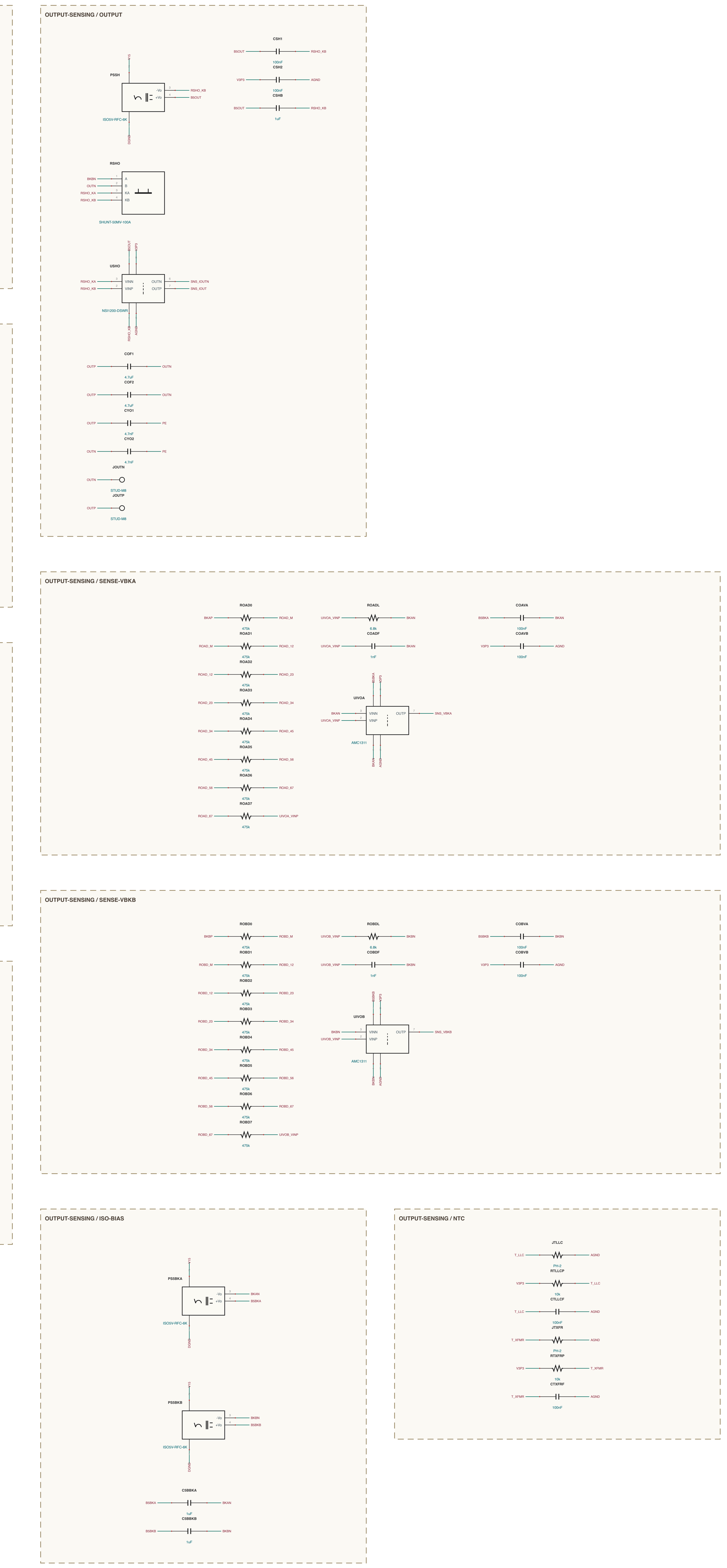










MAGNETICS CONSTRUCTION

identity per docs/magnetics.md (turns, gap, litz, acceptance lines live there)

D3 T1-T3: 3x PQ50/50 PC95 stack, 7:7:7 (E8), pri litz 2475x0.071 + sec Cu foil 0.10x28 (E60 Dowell rev), Lm 63uH +/-7% gap-ground

D2 L1T-L3T: gapped 2x PQ50/50, N=4, bins ~4.0uH, gap ground per bin

total Lr = trim + measured xfmr leakage (bins COMPENSATE leakage spread); Cr at TANK sections

SHEET INDEX

30 kW DC-DC (upper) - 2 of 3

OUTPUT-SENSING - 6 sections

BANKS-SP - 4 sections

LLC-LEGS - 4 sections

CONTROL - 3 sections

LLC-TANKS - 3 sections

COMMS-HMI - 2 sections

rev D.4 - 22 sections - 312 components

NET NAMING

internal junctions are named for what they join

U<ref>_<PIN> node at that IC pin e.g. UIVOA_VINP

+ 3 more

DC-Modules 30 kW — DC-DC board (3-phase LLC)

DC-Modules 30 kW - board DC-DC (upper), sheet 2 of 3

2026-09-06 · rev D.4